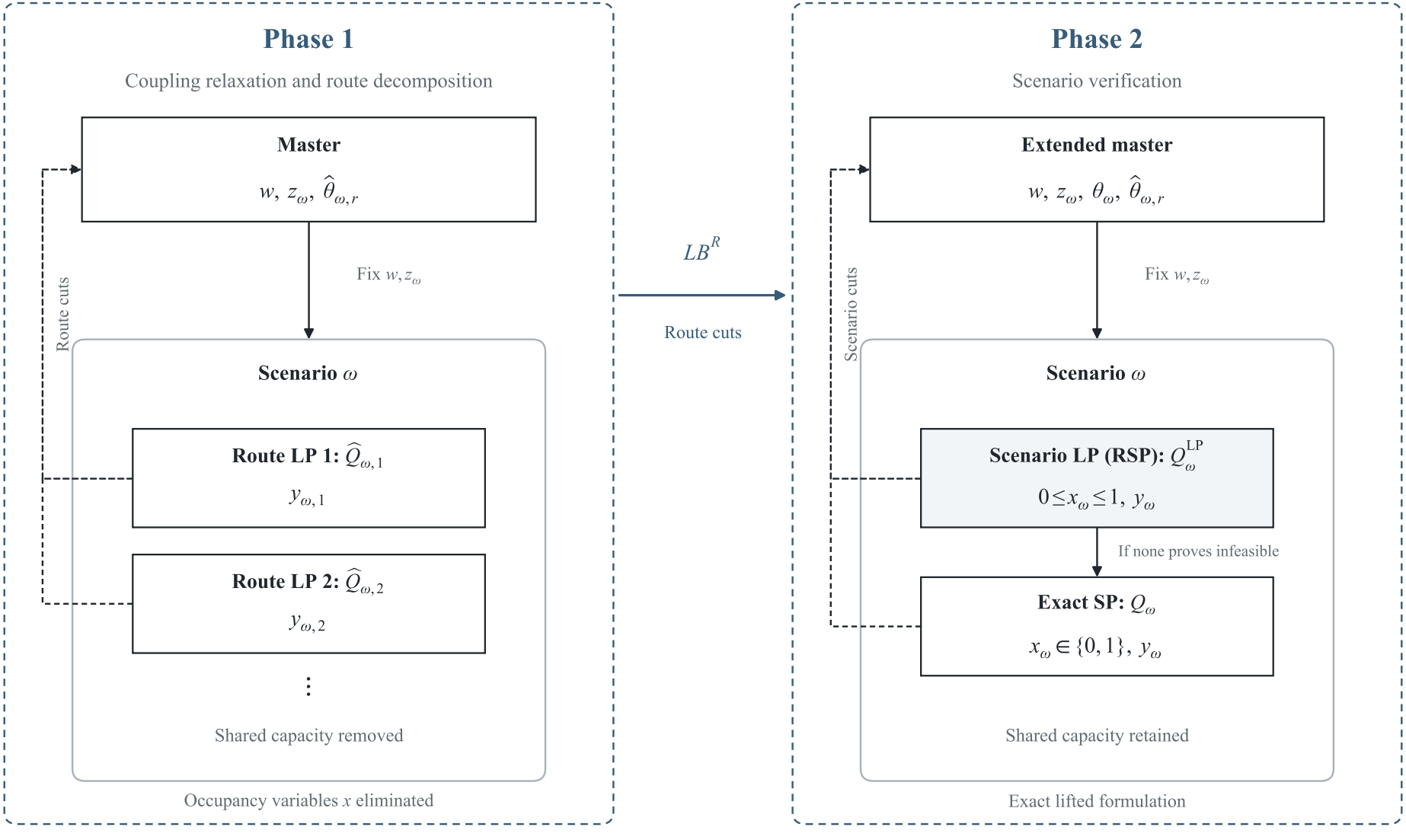
Phase 1
Coupling relaxation and route decomposition
Master
$w,\ z_\omega,\ \hat{\theta}_{\omega,r}$
Fix w, z_ω
Route cuts
Scenario ω
Route LP 1: $\widehat{Q}_{\omega,1}$
$y_{\omega,1}$
Route LP 2: $\widehat{Q}_{\omega,2}$
$y_{\omega,2}$
Shared capacity removed
Occupancy variables x eliminated
LB^R
Route cuts
Phase 2
Scenario verification
Extended master
$w,\ z_\omega,\ \theta_\omega,\ \hat{\theta}_{\omega,r}$
Fix w, z_ω
Scenario cuts
Scenario ω
Scenario LP (RSP): Q_ω^{LP}
$0 \le x_\omega \le 1,\ y_\omega$
If none proves infeasible
Exact SP: Q_ω
$x_\omega \in \{0,1\},\ y_\omega$
Shared capacity retained
Exact lifted formulation
RSP infeasible: return to the master.